UNIT 1 PRE-TEST

ANSWER KEY

The Real Number System

Answer each question, then check your work.

Name _____ Date _____ Period _____

SCORE *3.0* Basic Understanding

1.1 Converting Rational Numbers

Convert each number as directed. Write fractions in simplest form.

1. Write as a decimal: 3/8

= **0.375**

e.g. 0.5

Solution:

3 ÷ 8 = 0.375. It terminates, so it is rational.

2. Write as a fraction: 0.25

= **1/4**

simplest form, e.g. 1/4

Solution:

0.25 = 25/100 = 1/4 in simplest form.

1.2 Real Number Classification

Classify the number as Rational (R) or Irrational (IR).

3. Classify: √144

Classification: **R**

type R or IR

Solution:

√144 = 12, a whole number, so it is rational.

SCORE 4.0 Target Content

1.1 Converting Rational Numbers

Convert each number as directed. Write fractions in simplest form.

4. Write as a fraction: $0.\overline{6}$ (the 6 repeats)

= **2/3**

simplest form

Solution:

Let x = 0.666..., then 10x − x = 6, so 9x = 6 and x = 2/3.

5. Write as a decimal: $2\frac{5}{6}$ (2 and 5/6)

= **2.833...**

use ... for the repeating part

Solution:

2 5/6 = 2 + 5/6 = 2.8333... (the 3 repeats).

1.2 Real Number Classification

Classify the number as Rational (R) or Irrational (IR).

6. Classify: $\sqrt{20}$

Classification: **IR**

type R or IR

Solution:

20 is not a perfect square, so √20 = 2√5 ≈ 4.47... — non-terminating and non-repeating, so irrational.

Simplify each expression first, then classify the result.

7. Classify: $\sqrt{3} \times \sqrt{27}$

Classification: **R**

simplify, then R or IR

Solution:

√3 × √27 = √81 = 9, a whole number, so rational.

8. Classify: $\sqrt{32} \div \sqrt{2}$

Classification: **R**

simplify, then R or IR

Solution:

$\sqrt{32} \div \sqrt{2} = \sqrt{16} = 4$, a whole number, so rational.

1.3 Estimating Non-Perfect Square Roots

Estimate each value to the *nearest tenth* by clicking the number line.

9. Estimate: $\sqrt{40}$



Solution:

$6^2 = 36$ and $7^2 = 49$, so $\sqrt{40} \approx 6.32$, which rounds to 6.3.

10. Estimate: $\sqrt{95}$



Solution:

$9^2 = 81$ and $10^2 = 100$, so $\sqrt{95} \approx 9.75$, which rounds to 9.7.

SCORE *5.0* Advanced Content

1.2 Real Number Classification

Evaluate or simplify each expression if possible, then classify as Rational (R) or Irrational (IR).

11. Classify: $-(\sqrt{49} - \sqrt{36})$

Classification: **R**

simplify, then R or IR

Solution:

$-(\sqrt{49} - \sqrt{36}) = -(7 - 6) = -1$, an integer, so rational.

12. Classify: $3 + \sqrt{5}$

Classification: **IR**

type R or IR

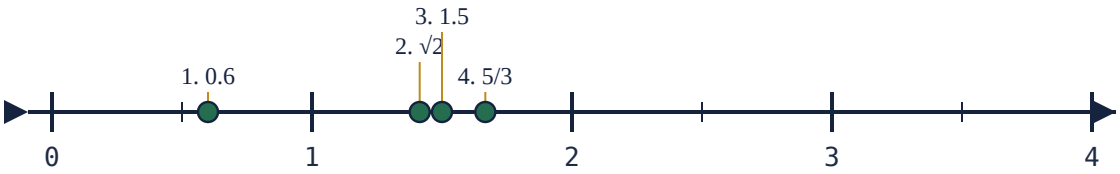
Solution:

√5 is irrational, and a rational number (3) plus an irrational number is irrational.

1.4 Ordering Real Numbers

Click the values in order from *least to greatest*. Each one drops onto the number line.

13. Set: { 0.6, √2, 1.5, 5/3 }



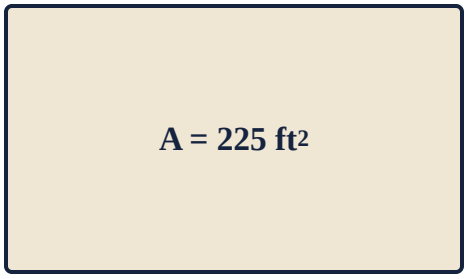
Order: **0.6, √2, 1.5, 5/3**

Solution:

As decimals: 0.6, √2 ≈ 1.41, 1.5, 5/3 ≈ 1.67 → least to greatest: 0.6, √2, 1.5, 5/3.

1.5 Area

A square garden has an area of 225 ft². Find the side length and the perimeter.



14. Find the side length and perimeter of the square garden.

Side length = **15 ft**

Perimeter = **60 ft**

Solution:

Side = √225 = 15 ft. Perimeter = 4 × 15 = 60 ft.